

MediLink: A Community Healthcare Information and Support Platform

Application of HCI Design Principles and HCI Guidelines
to the MediLink Project

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DE COSTA M.S.M.

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SENARATHNA S.A.D.H.D.

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Introduction

Healthcare services are essential for improving people's quality of life. However, many patients still face difficulties when accessing hospital services due to manual appointment booking, limited information about doctor and medicine availability and time-consuming payment processes. These challenges often lead to unnecessary hospital visits, long waiting times and inconvenience for patients.

To address these issues, we propose **MediLink**, a web-based healthcare platform designed to improve the patient experience through a simple and user-friendly interface. Based on the findings of our user requirements survey, the scope of the system focuses on three core patient services:

- Checking doctor availability and booking appointments.
- Checking medicine availability before visiting the hospital.
- Making secure online payments while providing patient details.

These features were selected because they directly address the main problems identified through the questionnaire survey. They also provide a complete and convenient user journey, allowing patients to check doctor availability, book appointments, verify medicine availability and complete payments online without unnecessary hospital visits. The system is designed following Human-Computer Interaction (HCI) principles to ensure usability, efficiency and a positive user experience.

MediLink at a Glance

MediLink was scoped, through a 103-response questionnaire, to three core patient services that respondents rated most highly: finding an available doctor and booking an appointment (88.3% support), checking medicine availability before travelling to hospital (84.5% support) and completing a secure online payment (76.7% willing to pay online). The primary users are patients (including elderly patients and parents/guardians booking on their behalf); doctors and hospital staff are supported as secondary, role-restricted users.

The system was designed around six core screens and a persistent, non-linear navigation model, implemented as a front-end prototype and finally connected to a working Node.js/Express/MongoDB back end. The screenshots used throughout this report are taken directly from that connected working system.

Technology Summary

Layer	Technology	Purpose
Front end	React + Vite	Patient-facing single-page application
Back end	Node.js + Express	REST API, validation, business rules
Database	MongoDB + Mongoose	Patients, doctors, hospitals, medicines, appointments, payments

Auth	JWT + bcrypt	Stateless authentication, hashed passwords
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Application of the Eight HCI Design Principles

Each of the eight principles below is stated as defined in the module material, followed by a concrete account of how MediLink applies it, illustrated with screenshots taken from the actual running prototype.

Know The User

Principle: design interaction and interfaces around the target users, using their age, computing experience and context to pick the right interaction model.

MediLink was designed for three distinct user roles identified with patients (including elderly patients and guardians booking for dependents), doctors and hospital staff; and the questionnaire data (68.9% of respondents aged 18-30, but a meaningful older minority) directly shaped the decision to keep the patient interface simple, high-contrast and touch-friendly rather than dense or command-driven. Each role is given its own dashboard, tailored to what that user needs to see and do, rather than a single generic screen shared by everyone.

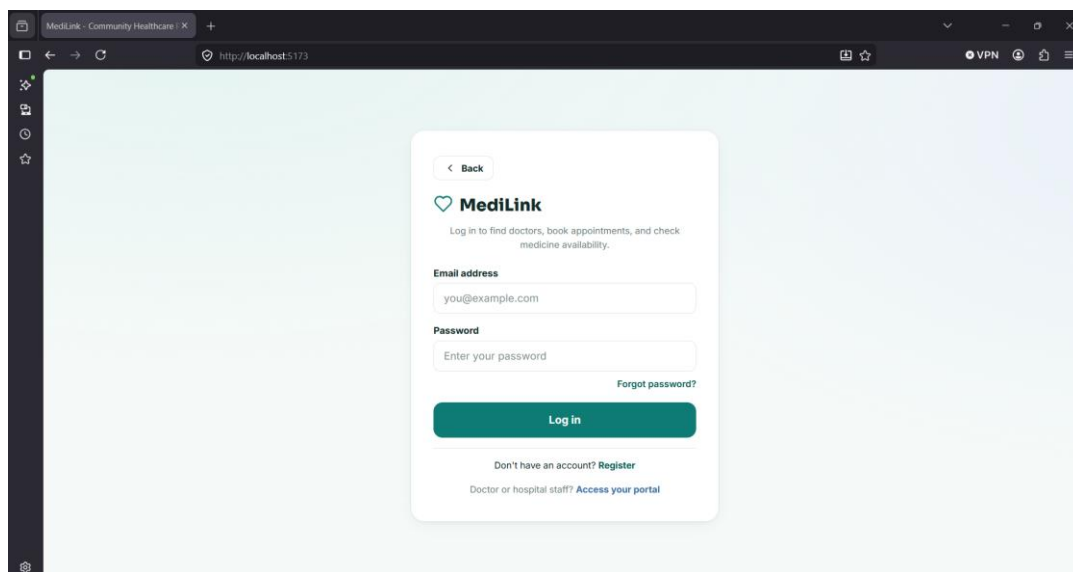


Figure 1: Login screen: a minimal, low-friction entry point suited to a mixed-skill patient population.

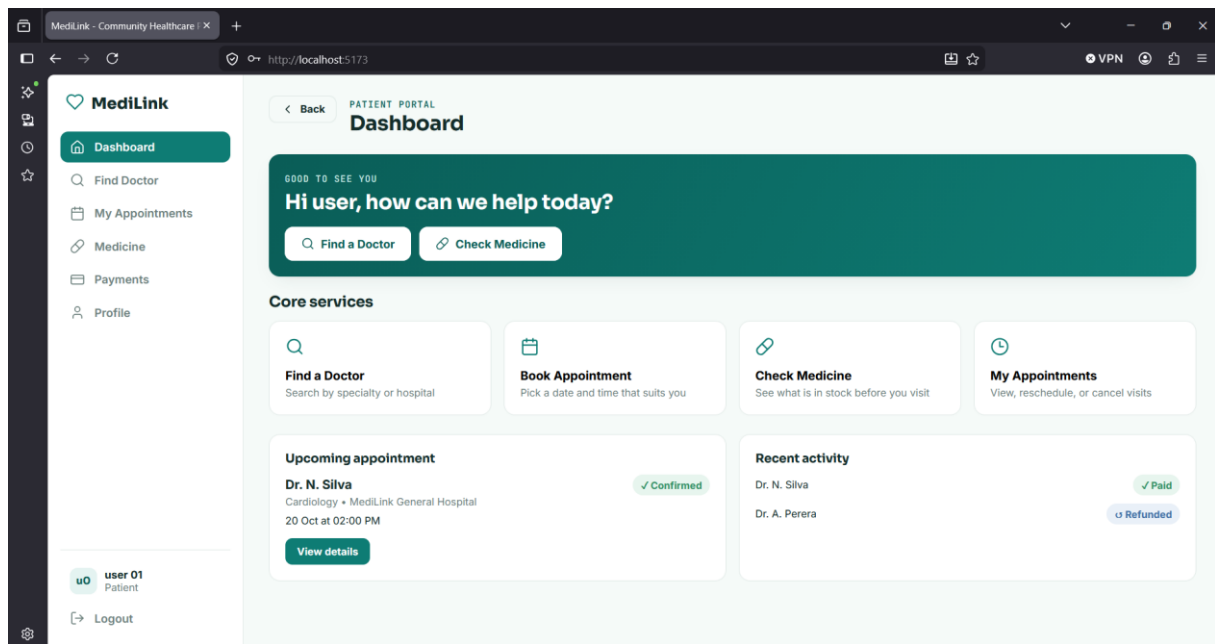


Figure 2: Patient dashboard: quick-action cards (Find a Doctor, Check Medicine, My Appointments) matched to the patient's core goal.

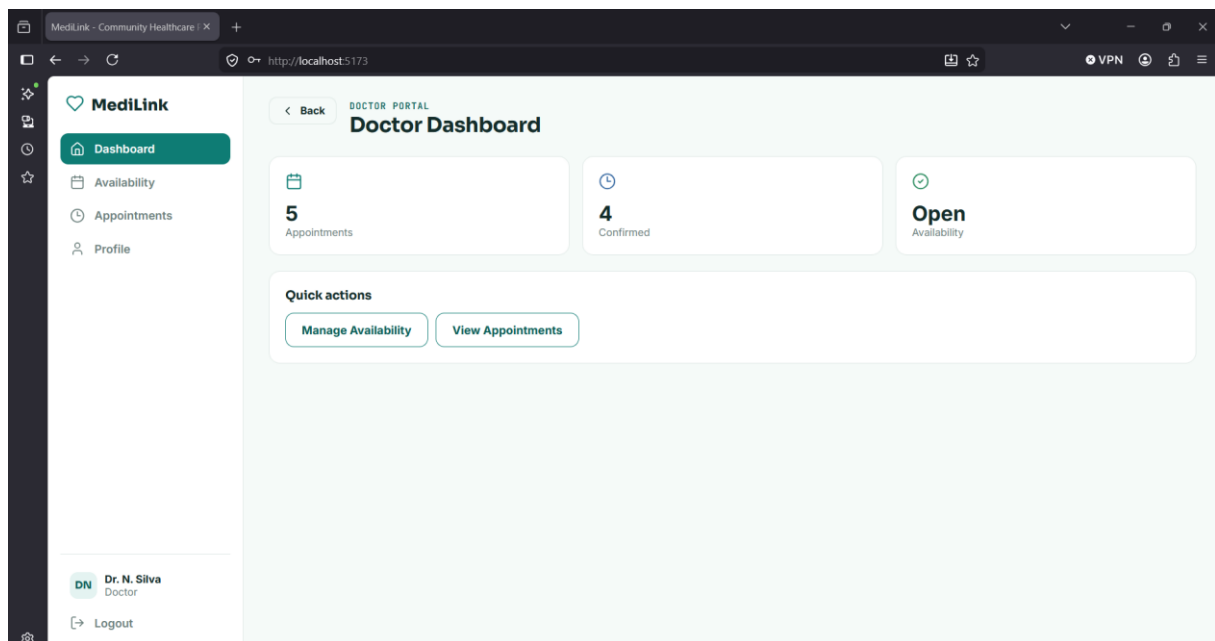


Figure 3: Doctor dashboard: a role-specific view for the doctor's own scheduling and patient-management needs.

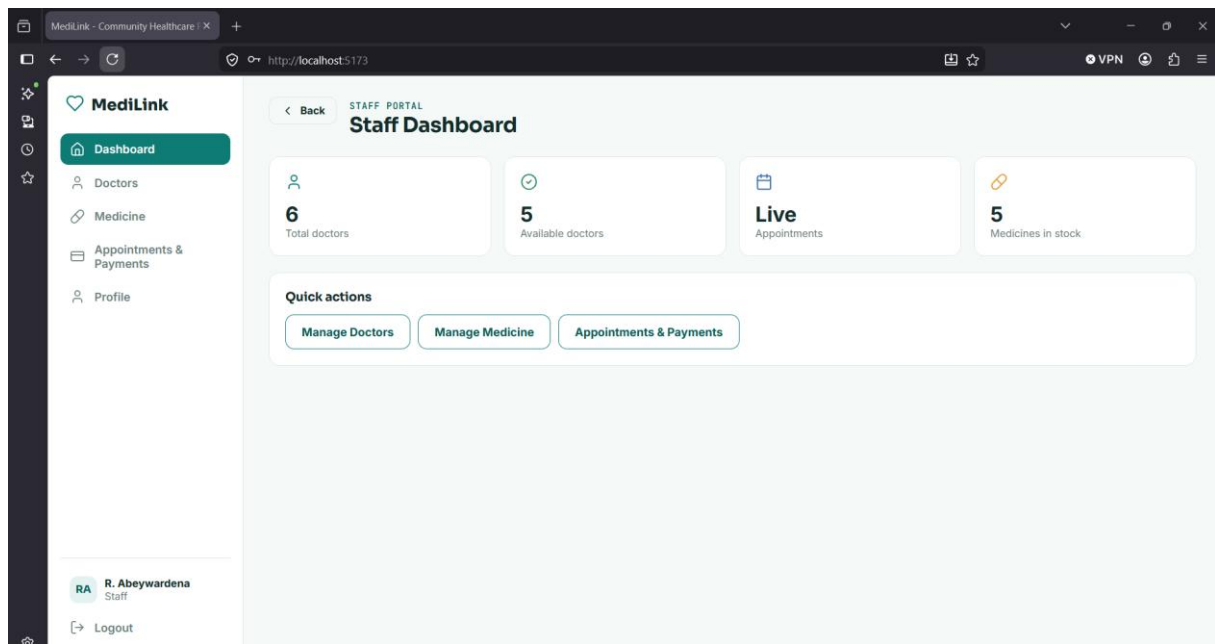


Figure 4: Hospital-staff dashboard: administrative view for managing doctors, medicine stock and payment verification, distinct from the patient view.

These three dashboards share the same visual language but present entirely different content and actions the same design decision the module material illustrates with the WhatsApp/Bloomberg Terminal and banking 'simple mode' examples: one backend, different interfaces for different users.

Universal Usability

Principle: serve a wide range of users across ages, skill levels and abilities, typically by offering separate or adaptable interfaces.

Universal usability is addressed at two levels. First, the patient registration and login screens require only the minimum fields needed to proceed, so first-time and infrequent users are not blocked by a long form, while returning or confident users move through quickly. Second, the same underlying platform is adapted for three different literacy/skill levels; the plain-language, icon-led patient dashboard; a denser doctor schedule view; and an administrative staff view; echoing the requirement that the interface remain usable by elderly patients (large touch targets, high-contrast text) without limiting what power users (doctors, staff) can do.

Figure 5: Registration screen: validated, minimal-field sign-up that supports guardians registering on behalf of a dependent.

The screenshot shows a web browser window with the URL `http://localhost:5173`. The page is titled "MediLink" and "PATIENT PORTAL Profile". On the left, there is a sidebar with navigation links: Dashboard, Find Doctor, My Appointments, Medicine, Payments, and Profile (which is highlighted). Below the sidebar, there is a user profile section for "user 01 Patient" with a "Logout" button. The main content area contains a "Profile" form with the following fields: "Full name" (containing "user 01"), "Email address" (containing "user01@gmail.com"), and "Phone number" (containing "0712345678"). At the bottom of the form are two buttons: "Save Changes" and "Logout".

Figure 6: Profile screen: lets any user, regardless of technical confidence, review and correct their own details after the fact.

Understanding the Task

Principle: identify the sequence and structure of sub-tasks at an abstraction level appropriate to the typical user, based on the user's own mental model.

The Hierarchical Task Analysis carried out and decomposed the patient's goal; 'secure a confirmed, paid appointment, optionally after checking medicine availability'; into four sub-tasks (login/register, find and book a doctor, check medicine, enter details and pay). The implemented interface mirrors that exact sequence screen-for-screen, so the interaction model matches the patient's own mental model of 'find a doctor, pick a time, pay' rather than exposing implementation detail (e.g. slot IDs, database identifiers) the patient does not think in terms of.

The screenshot shows a web browser window with the URL `http://localhost:5173`. The page is titled "MediLink" and "PATIENT PORTAL Find a Doctor". On the left, there is a sidebar with navigation links: Dashboard, Find Doctor (which is highlighted), My Appointments, Medicine, Payments, and Profile. Below the sidebar, there is a user profile section for "user 01 Patient" with a "Logout" button. The main content area contains a "Find a Doctor" form with the following elements: a search bar with the placeholder text "Search for doctors by name, specialty, or hospital, and see who is available now.", a "Search" button, and three dropdown menus labeled "Search", "Specialty", and "Hospital". Below the search bar, there is a section titled "Popular specialties" with five buttons: Cardiology, General Medicine, Dermatology, Pediatrics, and Orthopedics.

Figure 7: Find a doctor: search/filter by specialty and hospital

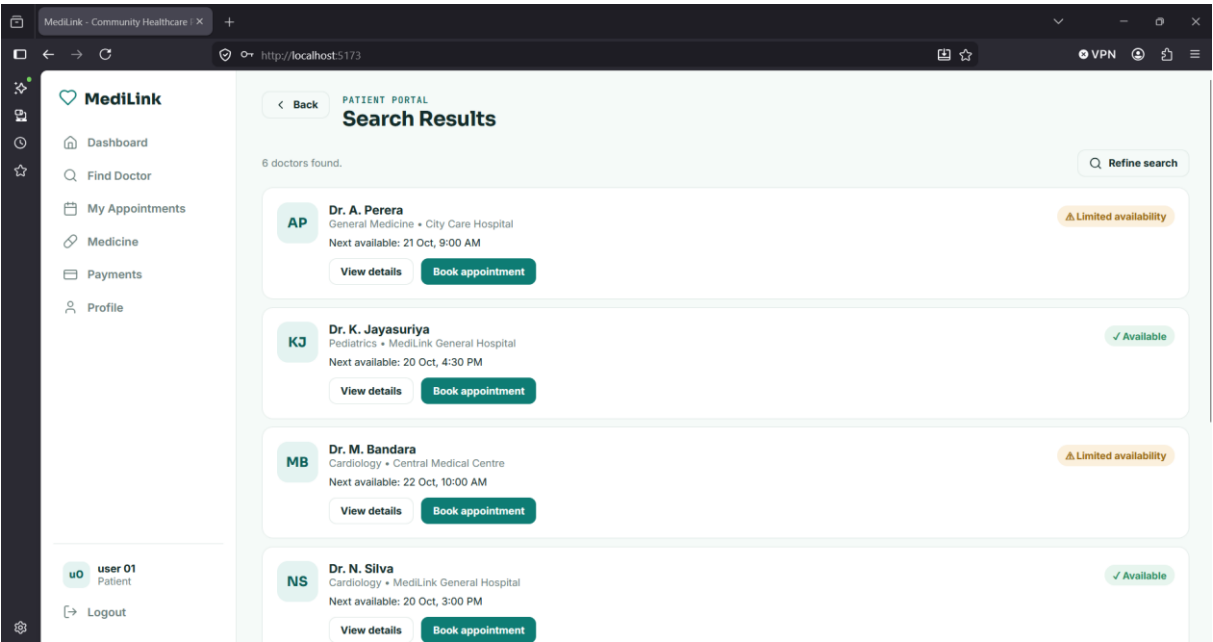


Figure 8: Search results: doctor cards with live availability badges, letting the patient choose without extra navigation

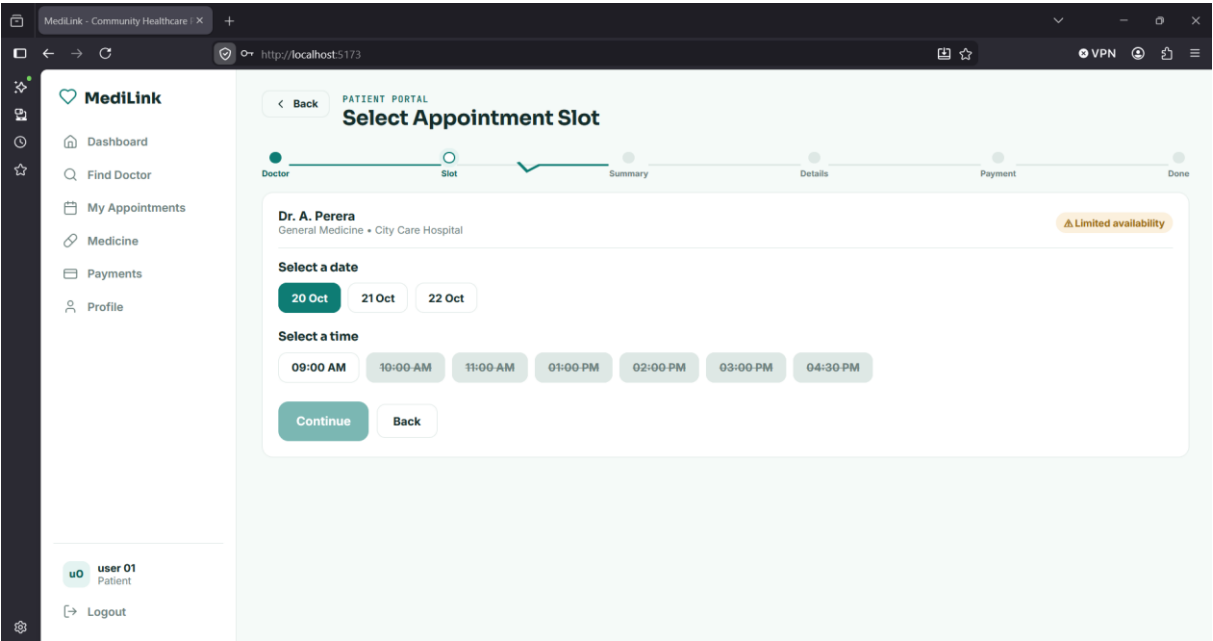


Figure 9: Appointment slot selection: date and time chips plus a stage tracker, matching the patient's step-by-step mental model of booking.

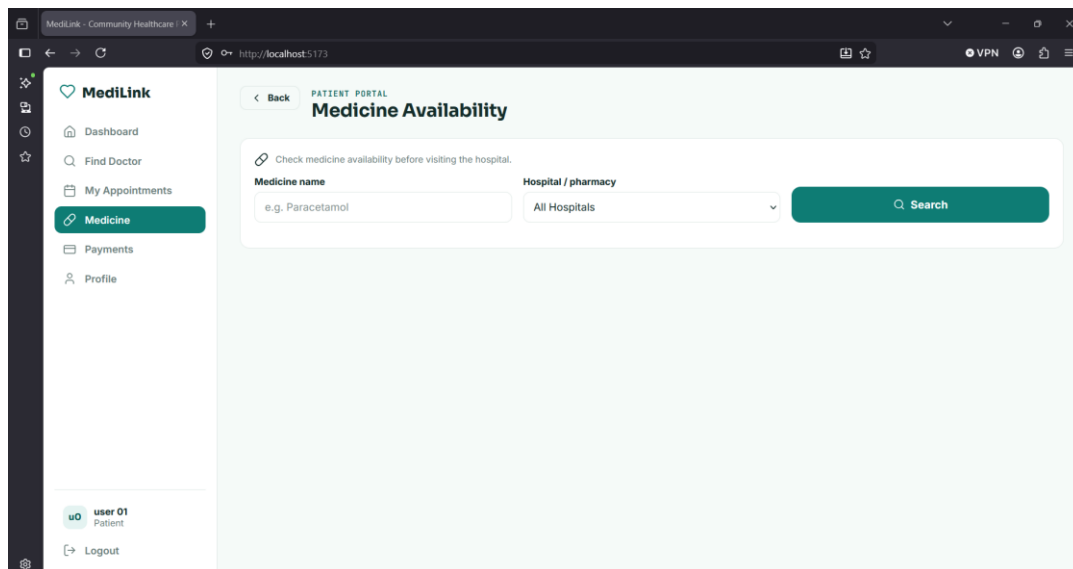


Figure 10: Medicine availability check: a self-contained sub-task the patient can perform independently of, or in the middle of, booking.

Reduce Memory Load

Principle: design so users must remember as little as possible; via chunking, recognition over recall and continuous status feedback.

Rather than asking the patient to recall the doctor, date, time and fee they selected several screens earlier, MediLink re-displays that information at every subsequent step. The Appointment Summary and Patient Details screens repeat the doctor, hospital, date, time and fee before any commitment is required, and the stage tracker at the top of the booking flow (Doctor → Slot → Summary → Details → Payment → Done) externalizes the patient's progress so they never have to hold 'which step am I on' in their head.

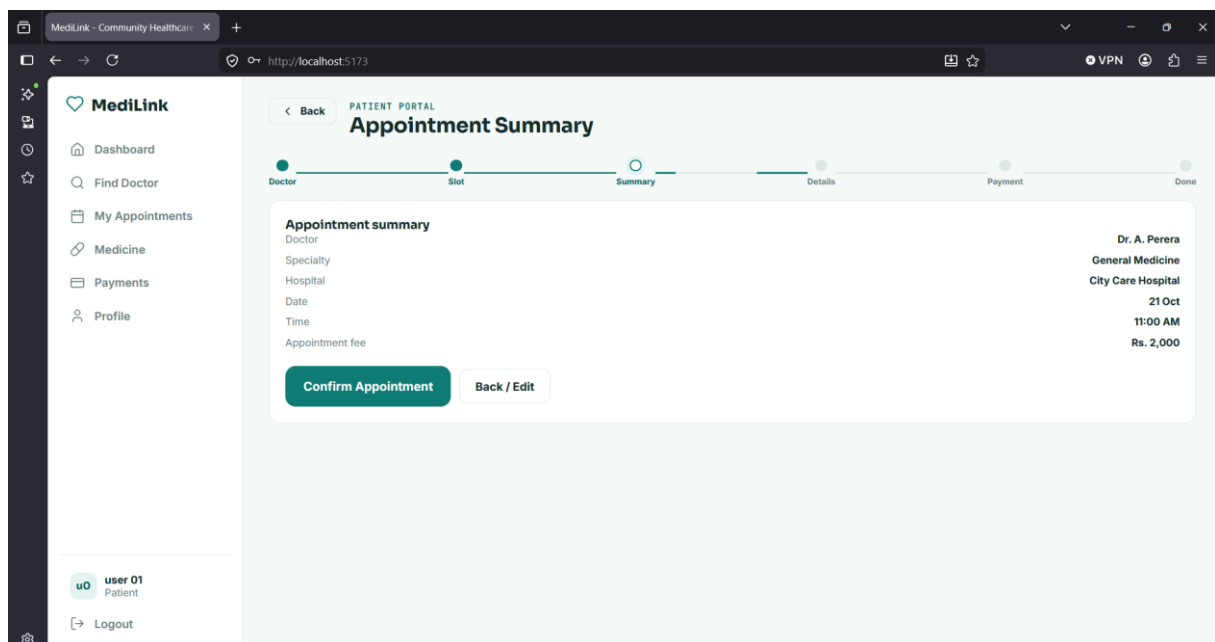


Figure 11: Appointment summary: doctor, specialty, hospital, date, time and fee are all re-displayed before the patient confirms nothing needs to be remembered from the previous screen.

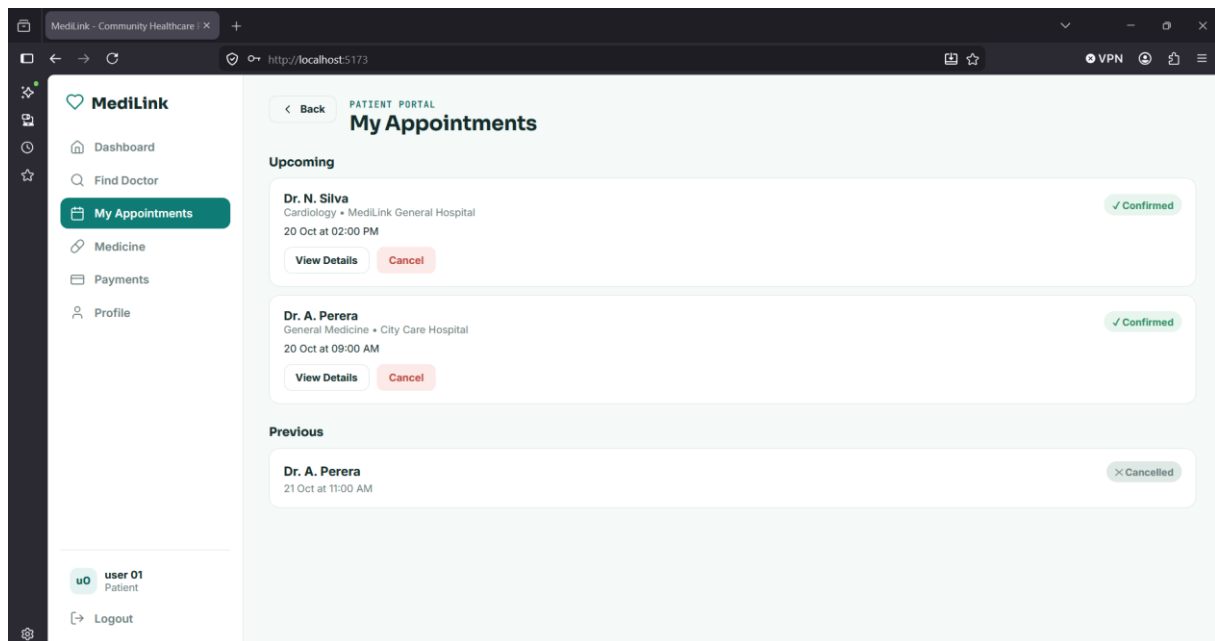


Figure 12: My Appointments: status badges (Confirmed / Completed / Cancelled) and a persistent list mean the patient never has to remember the state of a booking.

Strive for Consistency

Principle: what is learned once should work everywhere, in both interaction model and visual implementation.

A single design-token system (teal primary color, consistent card/button/badge styles, the same left-hand navigation structure) is reused across every patient screen and the same stage tracker, card layout and button placement recur through the entire booking journey. A patient who learns how to book an appointment has effectively already learned how to check medicine availability or review a payment, because the same controls (search fields, filter dropdowns, primary/secondary buttons, status badges) are reused rather than reinvented per screen.

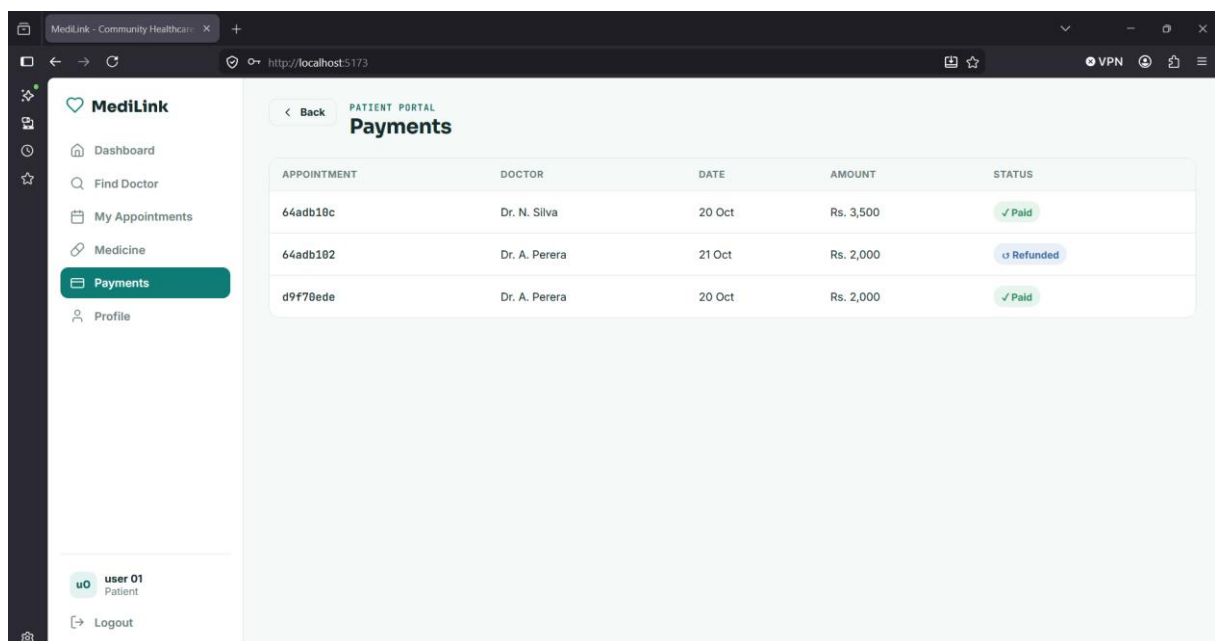


Figure 13: Payment method selection.

The screenshot shows the 'Payment' screen in the MediLink Patient Portal. The left sidebar contains navigation links: Dashboard, Find Doctor, My Appointments, Medicine, Payments, and Profile. The main content area has a progress bar with steps: Doctor, Slot, Summary, Details, Payment, and Done. The 'Payment method' section has two tabs: 'Card' (selected) and 'Online Banking'. Below the tabs are input fields for 'Cardholder name' (filled with 'user01'), 'Card number' (filled with '123456789'), 'Expiry date' (filled with '12/27'), and 'CVV' (filled with '...'). A security notice states: 'Your payment information is encrypted and protected. MediLink never stores your full card details.' A 'Pay Now' button is at the bottom. The 'Order summary' panel on the right displays: Doctor: Dr. A. Perera, Hospital: City Care Hospital, Date & time: 21 Oct, 11:00 AM, and Total: Rs. 2,000.

Figure 14: Card payment: the same summary-panel-plus-form layout used throughout the booking flow, just with a different form on the left.

This screenshot shows the 'Payment' screen with the 'Online Banking' payment method selected. The 'Payment method' section now has 'Online Banking' as the active tab. The form fields are: 'Select your bank' (a dropdown menu showing 'Commercial Bank'), 'Account holder name' (filled with 'user'), and a 'Pay Now' button. The 'Order summary' panel on the right remains identical to the previous screenshot, showing the same order details and total amount of Rs. 2,000.

Figure 15: Online banking option: switching payment method changes only the form fields; the surrounding layout, summary panel and button placement stay identical, reducing relearning.

Remind Users & Refresh Their Memory

Principle: keep reminding users of important information throughout a task, and give clear, reconfirming feedback when a larger process completes.

Before payment is submitted, MediLink surfaces a second confirmation summary so the patient re-checks the booking one last time; after payment, a dedicated confirmation screen restates the confirmation number, doctor, hospital, date/time, amount paid and payment status, giving the same kind of closure feedback the module material illustrates with ATM 'take your card' beeps and WhatsApp double ticks.

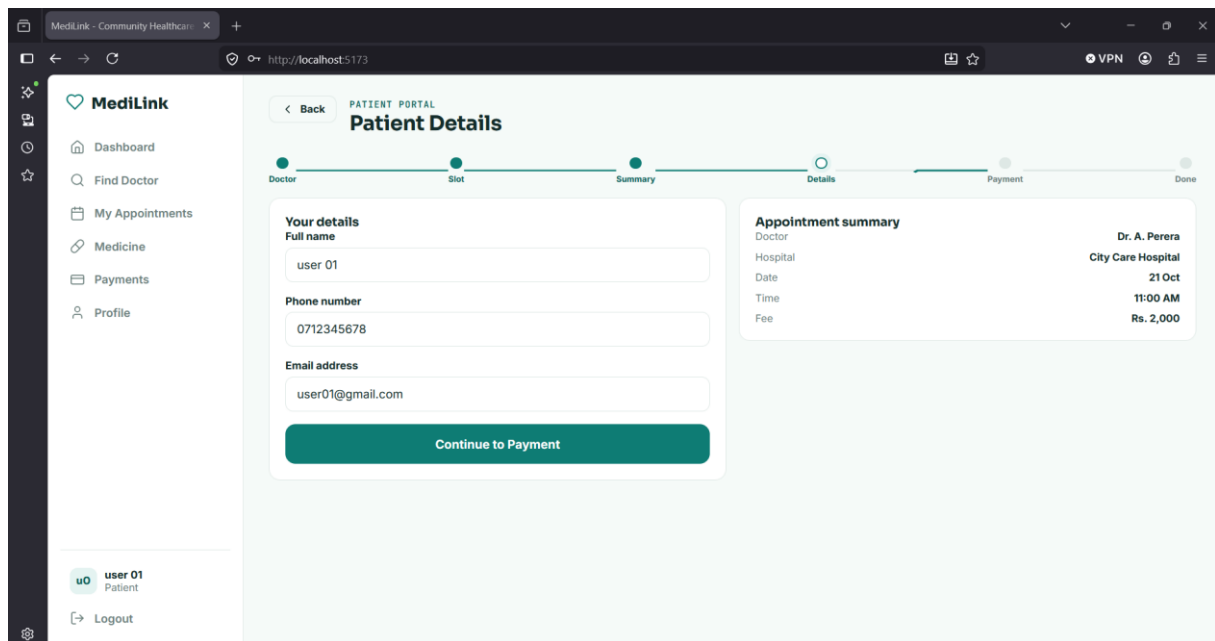


Figure 16: Pre-payment appointment review: the platform refreshes the patient's memory of exactly what they are about to pay for.

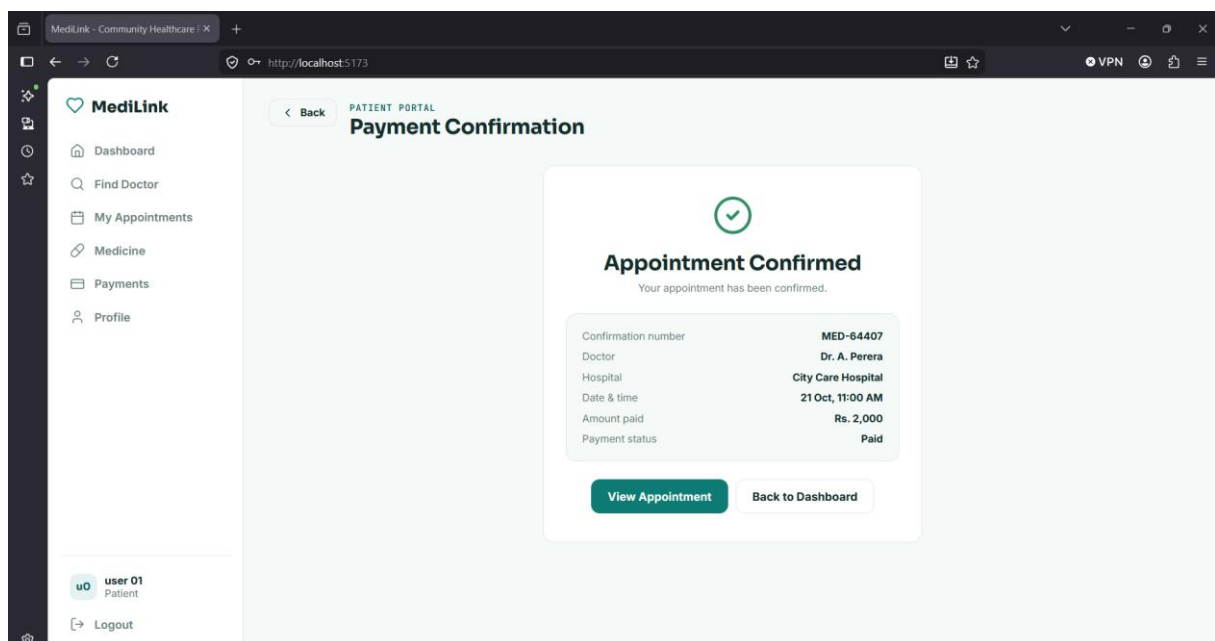


Figure 17: Payment confirmation: confirmation number, amount paid and status are shown together as closure feedback that the larger booking-and-payment process has succeeded.

Prevent Errors & Allow Reversal

Principle: constrain input to prevent mistakes, confirm risky/destructive actions, and always provide a way to reverse an action.

Unavailable doctor time slots are shown as disabled chips rather than hidden or selectable, so a patient physically cannot pick a slot that no longer exists; the interface prevents the error rather than catching it afterwards. On the back end, a database-level partial unique index prevents two patients from ever double-booking the same slot even under concurrent requests, converting any race condition into a clear, actionable 409 error rather than a silent double-booking. Destructive actions such as cancelling an appointment are behind a confirmation dialogue, and appointments can be rescheduled or cancelled after the fact; an explicit reversal path rather than a one-way commitment.

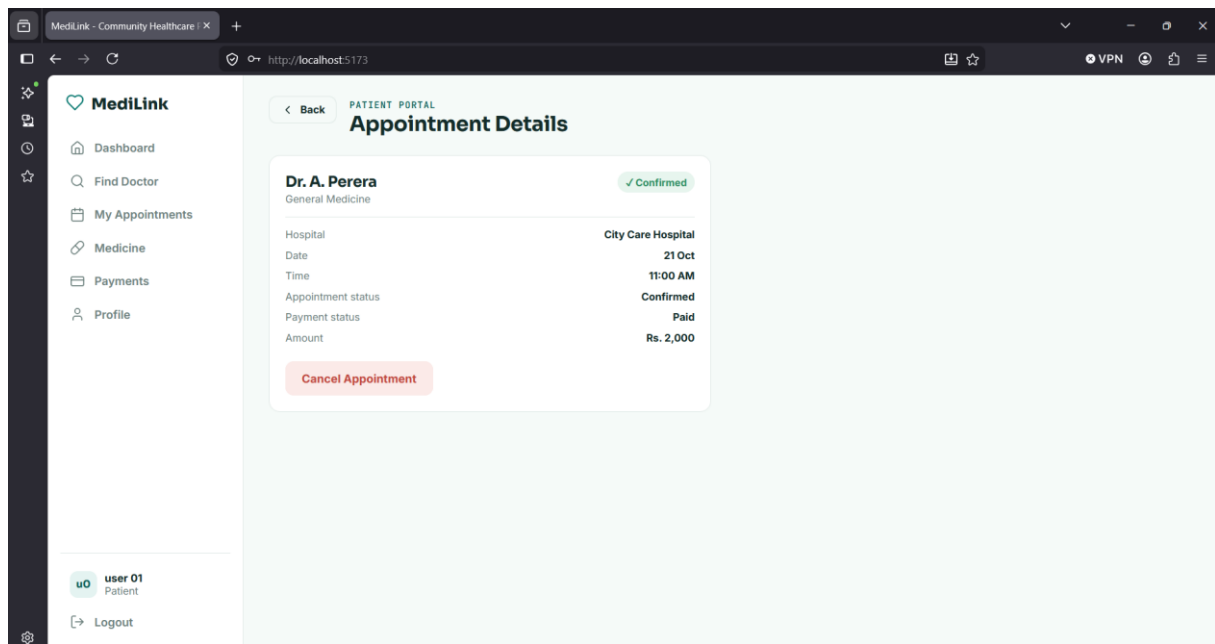


Figure 18: Final review before commitment: the patient can still go Back/Edit at this stage rather than being forced forward.

(My Appointments, Figure 12 above, is also the evidence screen for reversal: each upcoming appointment exposes Reschedule and Cancel actions and cancellation is gated behind a confirmation modal to avoid accidental loss of a booking.)

Naturalness

Principle: make interaction natural by reflecting everyday operations or by using an uncontrived metaphor that needs almost no learning.

MediLink favors interaction patterns patients already know from everyday apps rather than inventing new ones: date and time are picked as tappable chips (the same 'pick one from a row' pattern used by ride-hailing and food-delivery apps), payment method is chosen with a simple radio-style toggle between Card and Online Banking and the whole booking journey reads like the natural sequence of an in-person hospital visit; find a doctor, pick a time, give your details, pay; translated directly onto the screen rather than reorganized around database structure. No skeuomorphic page-turning or forced animation is used, keeping the metaphor honest rather than decorative.

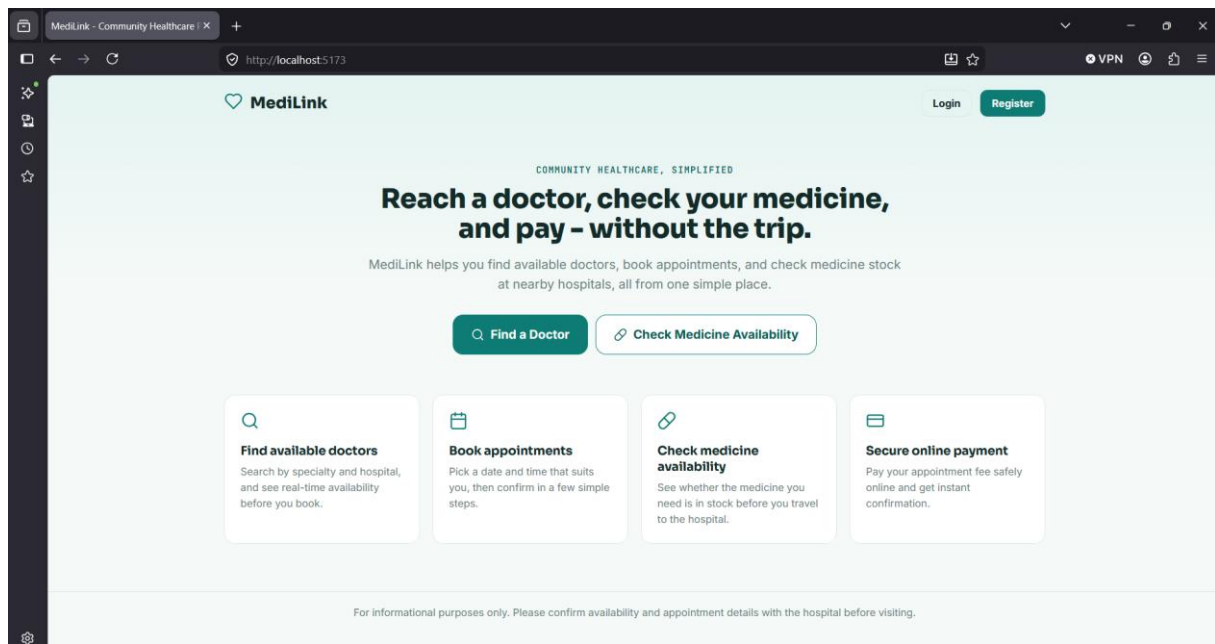


Figure 19: Landing page: plain-language service cards ('Find available doctors', 'Book appointments', 'Check medicine availability', 'Secure online payment') describe the natural, everyday tasks the platform performs, with no jargon or learning curve.

Application of HCI Guideline Criteria / Categories

Beyond the eight design principles, the module also supplies a broader classification of HCI guideline criteria (user type, platform/system setup, design style/interaction modality, task/operational context, and application domain). The table below positions MediLink within each category, with the corresponding screenshot evidence noted.

Category	Sub-criteria	How MediLink applies it
User type	Age/generation; Occupation; Disability/accessibility	Interface tuned for a mixed-age patient population (elderly-friendly high-contrast, large-target design) alongside occupation-specific roles; patient, doctor, hospital staff (Figures 2-4).
Platform / system setup	Desktop; Mobile/handheld	Fully responsive web app: persistent sidebar on desktop, collapsing to a mobile top bar, drawer and bottom navigation on small screens.
Design style / interface modality	WIMP (Windows, Icons, Menus, Pointer)	Classic WIMP interaction throughout; sidebar navigation, dropdown filters, card components, buttons and modal dialogues (Figures 1, 2, 7, 13) rather than voice, gesture or 3-D interaction.
Task / operational context	Home / general-purpose, time-of-day independent	Designed to be used from home, work or on the move (not tied to a hospital kiosk or fixed location), removing the need for an in-person or phone-based transaction.
Application domain	Electronic commerce / information & services	Combines an information-service application (doctor and medicine availability lookup) with a light e-commerce transaction (consultation-fee payment) in one coherent flow (Figures 13-17).

This mapping shows MediLink is not a single-category product: it deliberately spans several guideline categories at once (a general-purpose information-and-transaction platform, used across desktop and mobile, by three different user types, through a conventional WIMP interface); which is exactly the kind of multi-user, multi-context system the 'Know Thy User' and 'Universal Usability' principles in Section 3 were applied to address.

Screenshot Index

For quick cross-referencing, the table below lists every screenshot used in this report, the screen it depicts, and the principle(s) or guideline category it was used as evidence for.

Figure	Screen	Used as evidence for
1	Login	Know Thy User
2	Patient Dashboard	Know Thy User
3	Doctor Dashboard	Know Thy User
4	Hospital Staff Dashboard	Know Thy User
5	Register	Universal Usability
6	Profile	Universal Usability
7	Find a Doctor	Understand the Task
8	Find a Doctor - Results	Understand the Task
9	Select Appointment Slot	Understand the Task
10	Medicine Availability	Understand the Task
11	Appointment Summary	Reduce Memory Load
12	My Appointments	Reduce Memory Load; Prevent Errors & Allow Reversal
13	Payment (method selection)	Strive for Consistency
14	Card Payment	Strive for Consistency
15	Online Banking	Strive for Consistency
16	Appointment Summary (pre-payment)	Remind Users & Refresh Their Memory
17	Payment Confirmation	Remind Users & Refresh Their Memory
18	Appointment Summary (final review)	Prevent Errors & Allow Reversal
19	Landing Page	Naturalness

Conclusion

MediLink was built through a user-centered process; identifying target users and their needs, gathering and analyzing 103 survey responses, decomposing the patient's goal into a hierarchical task model, translating that model into a wireframed interface and finally implementing and connecting that interface to a working MERN-stack back end. The screenshots in this report demonstrate that the working prototype does not merely reference the eight HCI design principles and the broader guideline categories in its planning documents, but visibly embodies them in the interface a real user

interacts with: distinct, role-appropriate views for each user type; minimal-friction, accessible forms; a booking flow that mirrors the patient's own task model; persistent status and summary information that removes the need to remember; a single consistent visual and interaction language reused across every screen; explicit confirmation and reversal at every risky step; and natural, familiar interaction patterns throughout.

MediLink therefore provides both a documented design rationale and verifiable, running evidence that the system was designed and built in line with the HCI principles and guidelines set out for this module.